09 Sept 2020

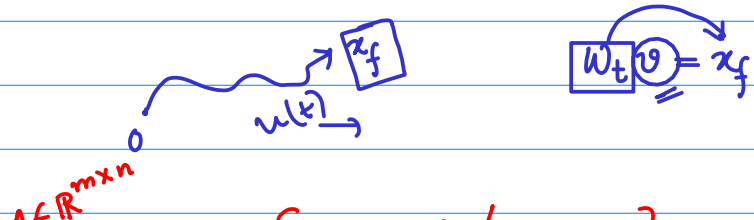
L2

For each time $t>0$, the following holds

$$\mathcal{R}_t = \operatorname{img} \mathcal{C}_{AB} = \operatorname{img} W_t$$

* $\mathcal{R}_t = \{\xi \in \mathbb{R}^n \mid \exists u \text{ such that } x(t) = \xi\}$
* $\mathcal{C}_{AB} = [B \;\; AB \;\; \cdots \;\; A^{n-1}B]$
* $W_t = \int_0^t e^{A\tau} B B^* e^{A^*\tau} d\tau$

Controllability Gramian



$$\left.\begin{aligned} \ker A &= \{x \in \mathbb{R}^n \mid Ax = 0\} \\ \operatorname{img} A &= \{y \in \mathbb{R}^m \mid y = Ax,\ x \in \mathbb{R}^n\} \end{aligned}\right] \rightarrow \text{Subspace}$$

Gilbert Strang

Let $T \in \mathbb{R}^{n\times n}$ be a non-singular matrix and $A \in \mathbb{R}^{n\times n}$, $B \in \mathbb{R}^{n\times m}$.
(A,B) is controllable $\Longleftrightarrow$ $(T^{-1}AT, T^{-1}B)$ is controllable.

Similarity transform.

<u>Proof:</u> Given (A,B) is controllable $\Rightarrow$ $\operatorname{rank}[B \;\; AB \;\cdots\; A^{n-1}B] = n$

To prove $(T^{-1}AT, T^{-1}B)$ is controllable

$T^{-1}B := \tilde{B} \Rightarrow B = T\tilde{B}$

$T^{-1}AT := \tilde{A} \Rightarrow A = T\tilde{A}T^{-1}$

$$\operatorname{rank}[\tilde{B} \;\; \tilde{A}\tilde{B} \;\cdots\; \tilde{A}^{n-1}\tilde{B}]$$

$$= \operatorname{rank}[T^{-1}B \;\; (T^{-1}AT)(T^{-1}B) \;\cdots\; (T^{-1}AT)^{n-1}(T^{-1}B)]$$

$$= \operatorname{rank}[T^{-1}B \;\; T^{-1}AB \;\cdots\; T^{-1}A^{n-1}B]$$

$$= \operatorname{rank}\left(T^{-1}[B \;\; AB \;\cdots\; A^{n-1}B]\right) \leq \min(\underbrace{\operatorname{rank}(T^{-1})}_{n}, \underbrace{\operatorname{rank}\mathcal{C}_{AB}}_{\bullet})$$

$$= \operatorname{rank}\mathcal{C}_{AB} = n$$

$$\operatorname{rank}\mathcal{C}_{AB} = \operatorname{rank}\mathcal{C}_{\tilde{A}\tilde{B}}$$

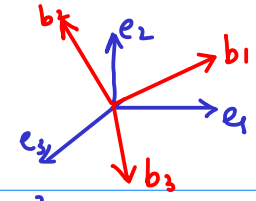
<u>Invariant Subspaces:</u> Let $\mathcal{W} \subseteq \mathbb{R}^n$ and $A \in \mathbb{R}^{n\times n}$. $\mathcal{W}$ is an A-invariant subspace if $A\mathcal{W} \subseteq \mathcal{W}$

$x \in \mathcal{W}$

$Ax = y \in \mathcal{W} \quad \forall x \in \mathcal{W}$

(P) $\mathcal{W} = \operatorname{img} w = \operatorname{img}\begin{pmatrix}1\\3\\4\end{pmatrix}$

1
3
4



$T = \begin{bmatrix} w & v_1 & v_2 \end{bmatrix}^{3\times 3} \longrightarrow$ non-singular

$x \in \mathbb{R}^3$

$\alpha\begin{pmatrix}1\\0\\0\end{pmatrix} + \beta\begin{pmatrix}0\\1\\0\end{pmatrix} + \gamma\begin{pmatrix}0\\0\\1\end{pmatrix}$

$$P\underbrace{\begin{bmatrix} w & v_1 & v_2 \end{bmatrix}}_{T} = \begin{bmatrix} \alpha w & Pv_1 & Pv_2 \end{bmatrix}$$

$1\underbrace{\begin{pmatrix}1\\0\\0\end{pmatrix}}_{e_1} + 3\underbrace{\begin{pmatrix}0\\1\\0\end{pmatrix}}_{e_2} + 4\underbrace{\begin{pmatrix}0\\0\\1\end{pmatrix}}_{e_3}$

$$= \begin{bmatrix} \alpha w & \beta_1 w + \beta_2 v_1 + \beta_3 v_2 & \gamma_1 w + \gamma_2 v_1 + \gamma_3 v_2 \end{bmatrix}$$

$\underset{w}{\begin{pmatrix}1\\3\\4\end{pmatrix}}$ $\underset{v_1}{\begin{pmatrix}1\\0\\0\end{pmatrix}}$ $\underset{v_2}{\begin{pmatrix}0\\1\\0\end{pmatrix}}$

$$= \underbrace{\begin{bmatrix} w & v_1 & v_2 \end{bmatrix}}_{T} \left[\begin{array}{c|cc} \alpha & \beta_1 & \gamma_1 \\ \hline 0 & \beta_2 & \gamma_2 \\ 0 & \beta_3 & \gamma_3 \end{array}\right]$$

P T non singular

$$\underline{\underline{T}}^{-1} P T = \left[\begin{array}{c|cc} \alpha & \beta_1 & \gamma_1 \\ \hline 0 & \beta_2 & \gamma_2 \\ 0 & \beta_3 & \gamma_3 \end{array}\right]$$

$\operatorname{img} \mathcal{C}_{AB} \longrightarrow$ Controllable space $\mathcal{W}$ $\quad (A, B)$ $\quad \dot{x} = Ax + Bu$

$\square$ $A \in \mathbb{R}^{n\times n}$, $B \in \mathbb{R}^{n\times m}$. Then $\operatorname{img} \mathcal{C}_{AB}$ is A-invariant.

<u>Proof:</u> $\mathcal{C}_{AB} = \begin{bmatrix} B & AB & A^2B & \cdots & A^{n-1}B \end{bmatrix}$

$y \in \operatorname{img} \mathcal{C}_{AB} \Rightarrow \exists v$ st $y = \begin{bmatrix} B & AB & A^2B & \cdots & A^{n-1}B \end{bmatrix} v$

$a_0 + a_1 s + \cdots a_{n-2} s^{n-2} = 0$

$A^2(a_0 I + a_1 A + \cdots a_{n-2} A^{n-2}) = 0$

$$Ay = A\begin{bmatrix} B & AB & A^2B & \cdots & A^{n-1}B \end{bmatrix} v$$

$$= \begin{bmatrix} AB & A^2B & A^3B & \cdots & A^{n-1}B & \underbrace{A^nB}_{\substack{\text{Linear comb}^n \text{ of} \\ B, AB \cdots A^{n-1}B}} \end{bmatrix} v$$

$a_0 + a_1 s + a_2 s^2 + \cdots a_{n-1} s^{n-1} + s^n \longrightarrow$ ch. pol. of A

$a_0 I + a_1 A + \cdots a_{n-1} A^{n-1} + A^n = 0 \Rightarrow A^n = -a_0 I - a_1 A - \cdots - a_{n-1} A^{n-1}$

$Ay = \text{linear comb.}(B, AB, \dots A^{n-1}B) \in \operatorname{img} \mathcal{C}_{AB}$

$A(\operatorname{img} \mathcal{C}_{AB}) \subseteq \operatorname{img} \mathcal{C}_{AB} \Rightarrow \operatorname{img} \mathcal{C}_{AB}$ is A-invariant

Given (A, B) and $\dim(\operatorname{img} \mathcal{C}_{AB}) = r \leq n$. Then $\exists$ a similarity transform $T \in \mathbb{R}^{n \times n}$ such that

ⓐ $T^{-1}AT = \begin{bmatrix} \tilde{A}_{11} & \tilde{A}_{12} \\ 0 & \tilde{A}_{22} \end{bmatrix}$ and $T^{-1}B = \begin{bmatrix} \tilde{B}_1 \\ 0 \end{bmatrix}$ where $\tilde{A}_{11} \in \mathbb{R}^{r \times r}$, $\tilde{A}_{22} \in \mathbb{R}^{(n-r) \times (n-r)}$, $\tilde{B}_1 \in \mathbb{R}^{r \times m}$

ⓑ $T(\operatorname{img} \mathcal{C}_{AB}) = \operatorname{img} \begin{bmatrix} I_r \\ 0 \end{bmatrix}$

ⓒ $(\tilde{A}_{11}, \tilde{B}_1)$ is controllable

$B = \underbrace{\begin{bmatrix} \overset{\nearrow w}{B} & v_1 & v_2 \end{bmatrix}}_{T} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

Ⓣ basis in $\mathcal{C}_{AB}$

$\begin{bmatrix} b_1 & b_2 & b_3 & \cdots \end{bmatrix}$ $\mathbb{R}^{n \times n}$

$T^{-1}B = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$